OS & SP

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SKILL – 20

1.Skill20.c

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <signal.h>
#include <sys/wait.h>
#include <termios.h>

#define MAX_ARGS 64

pid_t shell_pgid;
pid_t foreground_pgid = -1;
```

```
/* Ctrl+C handler */
void handle_sigint(int sig)
{
    (void)sig;

    if (foreground_pgid > 0) {
        kill(-foreground_pgid, SIGINT);
    }
}

/* Ctrl+Z handler */
void handle_sigtstp(int sig)
{
    (void)sig;

    if (foreground_pgid > 0) {
        kill(-foreground_pgid, SIGTSTP);
    }
```

```
}

int main(void)
{
    char command[256];

    /* Get shell PID */
    shell_pgid = getpid();

    /* Ignore terminal control signals in shell */
    signal(SIGTTOU, SIG_IGN);
    signal(SIGTTIN, SIG_IGN);
    signal(SIGTSTP, SIG_IGN);

    /* Create shell process group */
    if (setpgid(shell_pgid, shell_pgid) == -1 &&
        getpgrp() != shell_pgid) {
        perror("setpgid shell");
```

```
}

/* Give terminal to shell */
if (isatty(STDIN_FILENO)) {
   tcsetpgrp(STDIN_FILENO, shell_pgid);
}

/* Register signal handlers */
signal(SIGINT, handle_sigint);
signal(SIGTSTP, handle_sigtstp);

printf("====================================\n");
printf(" OSSP PROGRAM 20\n");
printf(" Process Groups and Terminal Control\n");
printf("====================================\n");

printf("Shell PID  : %d\n", shell_pgid);
printf("Shell PGID : %d\n", getpgrp());
```

```
printf("\nCommands:\n");
printf(" Any Linux command\n");
printf(" exit\n\n");

while (1) {
    printf("shell> ");
    fflush(stdout);

    if (fgets(command, sizeof(command), stdin) == NULL)
        break;

    command[strcspn(command, "\n")] = '\0';

    if (strlen(command) == 0)
        continue;

    if (strcmp(command, "exit") == 0)
```

```
    break;

/* Copy command before tokenizing */
char command_copy[256];

strcpy(command_copy, command);

/* Create argument array */
char *args[MAX_ARGS];
int argc = 0;

char *token = strtok(command_copy, " \t");

while (token != NULL && argc < MAX_ARGS - 1) {
    args[argc++] = token;
    token = strtok(NULL, " \t");
}
```

```
args[argc] = NULL;

if (argc == 0)
    continue;

/* Create child process */
pid_t pid = fork();

if (pid < 0) {
    perror("fork");
    continue;
}

/* CHILD */
if (pid == 0) {
    /* Create a new process group */
    setpgid(0, 0);
```

```
    /* Restore default signal behavior */
    signal(SIGINT, SIG_DFL);
    signal(SIGTSTP, SIG_DFL);
    signal(SIGTTIN, SIG_DFL);
    signal(SIGTTOU, SIG_DFL);

    /* Execute command */
    execvp(args[0], args);

    perror("execvp");
    _exit(1);
}

/* PARENT */
else {
    /* Put child in its own process group */
    setpgid(pid, pid);
```

```
foreground_pgid = pid;


printf("Child PID  : %d\n", pid);
printf("Child PGID : %d\n", getpgid(pid));
/* Transfer terminal control to child */
if (isatty(STDIN_FILENO)) {
   tcsetpgrp(STDIN_FILENO, foreground_pgid);
}
int status;
/* Wait for child to exit or stop */
if (waitpid(pid, &status, WUNTRACED) == -1) {
   perror("waitpid");
}
/* Return terminal control to shell */
if (isatty(STDIN_FILENO)) {
   tcsetpgrp(STDIN_FILENO, shell_pgid);
}
```

```
        foreground_pgid = -1;


        if (WIFEXITED(status)) {
            printf("Child exited normally.\n");
        }
        else if (WIFSIGNALED(status)) {
            printf("Child terminated by signal %d\n",
                   WTERMSIG(status));
        }
        else if (WIFSTOPPED(status)) {
            printf("Child stopped by signal %d\n",
                   WSTOPSIG(status));
            /* Resume stopped process, as in the source */
            kill(-pid, SIGCONT);
            waitpid(pid, NULL, 0);
        }
    }
}
```

```
    /* Restore terminal to shell */
    if (isatty(STDIN_FILENO)) {
        tcsetpgrp(STDIN_FILENO, shell_pgid);
    }
    printf("\nProgram 20 terminated.\n");
    return 0;
}
```

OUTPUT :

```
ramyasree@DESKTOP-DP2O9U6:~/Skill$ nano skill20.c
ramyasree@DESKTOP-DP2O9U6:~/Skill$ gcc -Wall -Wextra -g skill20.c -o skill20
ramyasree@DESKTOP-DP2O9U6:~/Skill$ ./skill20
====================================
 OSSP PROGRAM 20
 Process Groups and Terminal Control
====================================
Shell PID  : 2307
Shell PGID : 2307

Commands:
 Any Linux command
 exit

shell> pwd
Child PID  : 2308
Child PGID : 2308
/home/ramyasree/Skill
Child exited normally.
shell> ls
Child PID  : 2309
Child PGID : 2309
append.txt    skill10    skill12.c  skill15    skill17.c  skill2     skill3.c         skill5    skill7.c
combined.txt  skill10.c  skill13    skill15.c  skill18    skill2.c   skill3dynamic    skill5.c  skill8
errors.txt    skill11    skill13.c  skill16    skill18.c  skill20    skill3dynamic.c  skill6    skill8.c
input.txt     skill11.c  skill14    skill16.c  skill19    skill20.c  skill4           skill6.c  skill9
output.txt    skill12    skill14.c  skill17    skill19.c  skill3     skill4.c         skill7    skill9.c
Child exited normally.
shell>
```